Bond Breaking & Bond Forming

Exam code: 0620 & 0971

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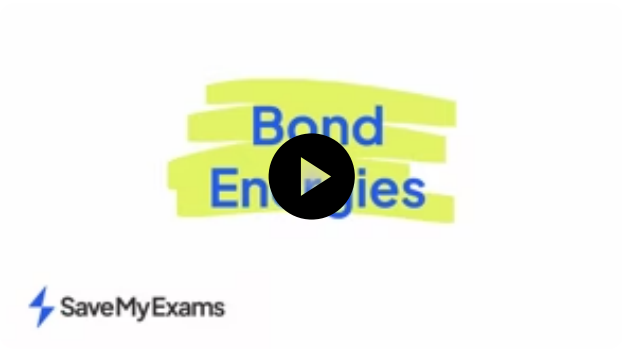

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Bond breaking & bond forming

Extended tier only

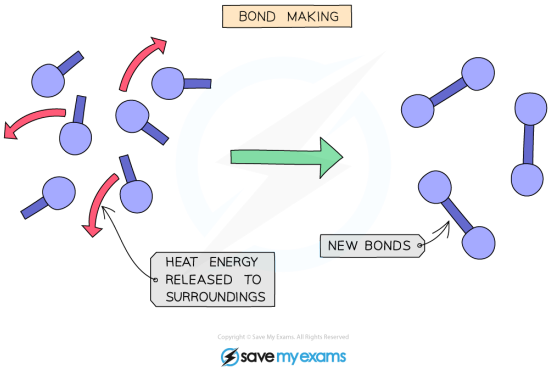
- Whether a reaction is endothermic or exothermic depends on the difference between the energy needed to **break** existing bonds and the energy released when the new bonds are **formed**
- **Bond breaking** is always an **endothermic** process as energy needs to be taken in from the surroundings to break the chemical bonds
- **Bond making** is always an **exothermic** process as energy is transferred to the surroundings as the new bond is formed

Exothermic reactions

- If more energy is released than is absorbed, then the reaction is **exothermic**
- More energy is released when new bonds are formed than energy required to break the bonds in the reactants
- The change in energy is negative since the products have less energy than the reactants
- Therefore, an exothermic reaction has a **negative** ΔH value

Help me understand

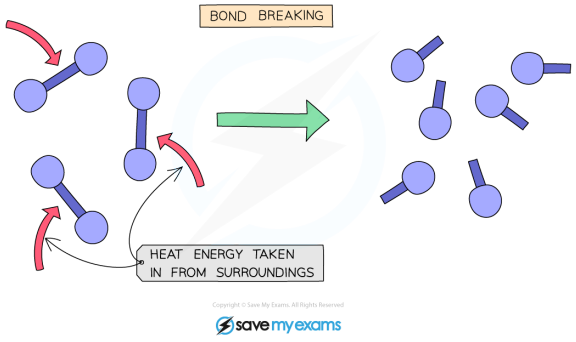
Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



Making new chemical bonds releases energy which radiates outwards from the reaction to the surroundings in the form of heat

Endothermic reactions

- If more energy is absorbed to break bonds than is released to form new bonds, this reaction is **endothermic overall**
- The change in energy is positive since the products have more energy than the reactants
- The symbol ΔH is used to show the change in heat energy
  - H is the symbol for enthaply, which is a measure of the total **heat of reaction** of a chemical reaction
- Therefore, an endothermic reaction has a **positive** ΔH value
  - This can be shown in reaction pathway diagrams and calculations



Breaking chemical bonds requires energy which is taken in from the surroundings in the form of heat

Bond energy calculations

Extended tier only

- Each chemical bond has specific bond energy associated with it

would be released or absorbed in a reaction
- To do this it is necessary to know the bonds present in both the reactants and products

How to complete bond energy calculations

- Write a balanced equation if none is present already
- Optional - draw the displayed formula in order to identify the type and number of bonds more easily
- Add together all the bond energies for all the bonds in the reactants – this is the 'energy in'
- Add together the bond energies for all the bonds in the products – this is the 'energy out'
- Calculate the enthalpy change:

Enthalpy change (ΔH) = Energy taken in – Energy given out

Worked Example

Hydrogen and chlorine react to form hydrogen chloride gas:

$$H_2 + Cl_2 \rightarrow 2HCl$$

The bond energies are given in the table below.

Bond	Energy (kJ)
H–H	436
Cl–Cl	242
H–Cl	431

Calculate the overall energy change for this reaction and use this value to explain whether the reaction is exothermic or endothermic.

Answer:

- Calculate the energy in
  - 436 + 242 = 678 (kJ)
- Calculate the energy out
  - 2 x 431 = 862 (kJ)
- Calculate the energy change
  - 678 - 862 = **–184** (kJ)
- Since the energy change is a negative number, energy is being released (to the surroundings)
  - Therefore, the reaction is **exothermic**

Examiner Tips and Tricks

When calculating enthalpy change using bond energies, it is helpful to write down a **displayed formula** equation for the

So, the reaction for the above worked example is:

$$H-H + Cl-Cl \rightarrow H-Cl + H-Cl$$

Worked Example

Hydrogen reacts with iodine to form hydrogen iodide.

$$H_2 + I_2 \rightarrow 2HI$$

The relevant bond energies are shown in the table below.

Bond	Energy (kJ)
H–I	295
H–H	436
I–I	151

Calculate the overall energy change for this reaction and use this value to explain why the reaction is exothermic.

Answer:

- Calculate the energy in
  - 436 + 151 = 587 (kJ)
- Calculate the energy out
  - 2 x 295 = 590 (kJ)
- Calculate the energy change
  - 587 - 590 = **-3** (kJ)
- The reaction is exothermic because:
  - **More energy is released than taken in**

Worked Example

Hydrogen bromide decomposes to form hydrogen and bromine:

$$2HBr \rightarrow H_2 + Br_2$$

The overall energy change for this reaction is +103 kJ.

The relevant bond energies are shown in the table below.

Bond	Energy (kJ)
H–Br	366
Br–Br	
H–H	436

Calculate the bond energy of the Br–Br bond.

Answer:

- 436 + Br–Br
- Overall energy change = energy in - energy out
  - +103 = 732 - (436 + Br–Br)
  - +103 = 732 - 436 - Br–Br
- Calculate the bond energy of the Br–Br bond
  - Br–Br = 732 - 436 - 103
  - Br–Br = +**193 (kJ)**

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